Validation Boundaries of Reduced-Order Inverse Protection Models Under Sixth-Order Synchronous Machine Dynamics

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***Abstract*—Reduced-order inverse models are attractive for real-time adaptive overcurrent protection because they identify relay-relevant source parameters within a protection operating interval. Their reliability, however, depends on the degree to which the true generator fault response remains representable by the reduced waveform model over the estimation window. This paper develops a physics-faithful forward benchmark based on a sixth-order synchronous-machine Park dq model with automatic voltage regulator (AVR), power system stabilizer (PSS), and governor dynamics, and uses it to quantify the validity limits of the Stage-1 reduced-order inverse protection model. The forward benchmark is formulated as a nine-state stiff ordinary differential equation (ODE) system, coupled to a stator algebraic interface and inverse Park transformation to generate relay-observable three-phase currents. Window-dependent model mismatch is evaluated using four complementary metrics: waveform residual norm, parameter bias, normalised Jacobian condition number, and confidence-gate behaviour. Results show that the reduced model remains relay-valid for estimation windows up to approximately 150 ms, achieving waveform residuals below 0.05 p.u. and confidence scores above the acceptance threshold $\gamma_{th} = 0.70$. Beyond this boundary, AVR-driven evolution of the sustained current component progressively biases the estimated steady-state current $\hat{I}_{ss}$ downward and inflates the estimated AC time constant $\hat{\tau}_{ac}$, causing pickup errors exceeding 10% and trip-time deviations exceeding the grading margin requirement. A relay-validity map is presented that delineates the window-length and operating-point region over which reduced inverse adaptation may be trusted, cautioned, or blocked. The paper thus provides a protection-oriented identifiability and fidelity framework for the deployment of reduced inverse estimators in adaptive relay IEDs.**

***Index Terms*—adaptive overcurrent protection, synchronous generator, Park dq model, sixth-order machine model, inverse estimation, window validity, Levenberg–Marquardt, condition number, confidence gate, AVR, PSS, governor.**

I. Introduction

A. Motivation

Online parameter estimation offers a principled route to adaptive overcurrent protection: rather than selecting from a discrete table of pre-computed settings, the relay fits a parametric fault-current model to the observed waveform and maps the result directly to protective relay settings [1], [2]. The Stage-1 SAMBP-SyncOC framework demonstrated that a six-parameter reduced inverse model, fitted by a two-pass Levenberg–Marquardt (LM) estimator, can extract relay-relevant quantities within the first few cycles after fault inception [1].

The reduced model captures the dominant waveform features: an AC transient driven by the direct-axis transient time constant, a DC offset decaying with the armature circuit time constant, and a steady-state sinusoidal component whose magnitude approximates the post-fault rms current. These features are fully present within the first 100–150 ms of a fault.

Over longer windows, however, the true generator response is shaped by controller dynamics that the reduced model deliberately omits:

- The AVR boosts field voltage E_{fd} within ∼0.1 s, causing the sustained current I_{ss} to rise progressively after the initial subtransient period.
- The governor adjusts mechanical torque T_m on a ∼0.5 s time scale, modifying the air-gap torque balance and rotor speed.
- The PSS injects a damping signal into the AVR to suppress rotor oscillations, creating superimposed oscillatory components in the fault current.

None of these dynamics appear in the six-parameter reduced model. When the estimation window extends into the range where these dynamics are active, the reduced model is misspecified: the LM estimator minimises residuals by distorting the inferred parameters, introducing systematic bias that may degrade relay decisions.

B. Gap in the literature

Prior work on adaptive overcurrent protection addresses estimator design [2], [3] or relay logic [4], [5], but rarely asks the foundational question: *over what window is the reduced model a sufficiently faithful representation of the true fault current for adaptive relay decisions*?

Physics-based synchronous-machine models of the sixth-order Park dq form are well-established in power system stability analysis [6], [7]. They have not, however, been systematically employed as truth-model benchmarks for evaluating relay-relevant identifiability boundaries. This paper bridges that gap.

Manuscript received April 9, 2026. This work establishes the physics-faithful truth-model validation framework (Stage-2) for the SAMBP-SyncOC adaptive protection series and is part of the PhD research programme at the Department of Electrical Engineering, Indian Institute of Technology Madras, Chennai 600 036, India.

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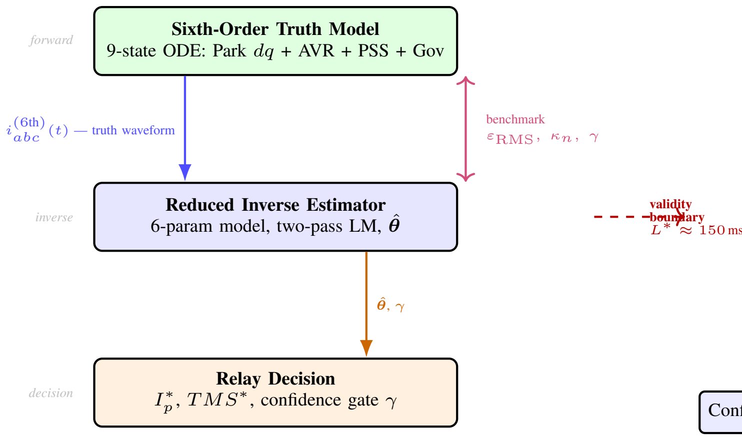

Fig. 1. Paper framework: the sixth-order truth model generates relay-observable fault currents that benchmark the reduced inverse estimator. The validity boundary defines where the estimator remains accurate enough for adaptive relay decisions.

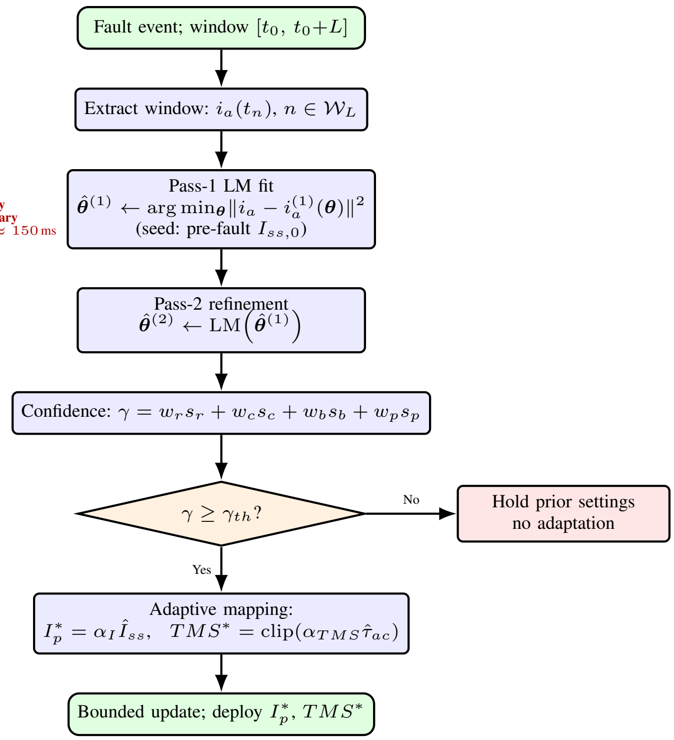

Fig. 2. Stage-1 reduced inverse estimator: two-pass Levenberg–Marquardt fit followed by confidence gating and adaptive relay mapping.

C. Paper objective

The central objective is to *quantify the relay-validity boundary of the reduced inverse protection model under sixth-order synchronous-machine dynamics with AVR, PSS, and governor*. The sixth-order model is not proposed as an alternative estimator — it is computationally prohibitive for real-time IED use — but as a physics-faithful ground truth against which the simpler model's accuracy is measured. Fig. 1 illustrates the three-layer architecture of this study.

D. Contributions

The main contributions are as follows.

1) A physics-faithful sixth-order Park dq forward benchmark for synchronous-generator fault-current simulation, augmented with AVR, PSS, and governor dynamics, coupled to a stator algebraic interface and inverse Park transformation to generate relay-observable abc currents (Section III).
2) A protection-oriented validation methodology comparing the reduced inverse model against the sixth-order benchmark across window lengths, fault types, and operating conditions, using four relay-relevant metrics (Section IV).
3) A window-validity analysis identifying the time horizon up to which the reduced inverse model is relay valid ($L^* \approx 150$ ms), and beyond which neglected AVR/PSS/governor dynamics materially bias estimated parameters (Section VI).
4) An identifiability and confidence study linking residual growth to Jacobian conditioning, parameter bias, and confidence-gate behaviour in the Stage-1 inverse estimator (Section VI).
5) A relay-validity map and practical deployment guidelines specifying when reduced inverse adaptation may be trusted, flagged, or blocked (Sections VI and VII).

II. Baseline Reduced-Order Inverse Protection Model

A. Analytical waveform model

The Stage-1 reduced inverse model represents the phase-A fault current as [1]

$$i_a^{(1)}(t) = \left[I_{ss}+(I_{sub}-I_{ss})\, e^{-t_{\text{rel}}/\tau_{ac}}\right] \sin(\omega t+\varphi_a) - I_{sub} \sin(\varphi_a)\, e^{-t_{\text{rel}}/\tau_{dc}} \tag{1}$$

where $t_{\text{rel}} = t - t_0$, I_{ss} is the sustained (steady-state) current magnitude, I_{sub} is the subtransient peak, τ_{ac} is the AC decay time constant (approximately T'_{d0}), τ_{dc} is the DC offset decay constant, and φ_a is the fault inception angle for phase A. The term $-I_{sub} \sin(\varphi_a)\, e^{-t/\tau_{dc}}$ provides the maximum possible DC offset consistent with $i_a(t_0) = 0$ [7].

The six-parameter vector is

$$\boldsymbol{\theta} = \left[t_0,\ I_{ss},\ I_{sub},\ \tau_{ac},\ \tau_{dc},\ \varphi_a\right]^\top. \tag{2}$$

B. Two-pass Levenberg–Marquardt estimator

Pass 1 seeds I_{ss} from the pre-fault rms current and minimises

$$\hat{\boldsymbol{\theta}}^{(1)} = \arg\min_{\boldsymbol{\theta}} \sum_{n \in \mathcal{W}} \left[i_a(t_n) - i_a^{(1)}(\boldsymbol{\theta}, t_n)\right]^2, \tag{3}$$

over window $\mathcal{W} = \{t_n : t_0 \leq t_n \leq t_0 + L\}$. Pass 2 uses $\hat{\boldsymbol{\theta}}^{(1)}$ as a warm start and refines the estimate. Fig. 2 shows the estimator architecture.

TABLE I
BASELINE REDUCED INVERSE MODEL: PARAMETERS AND ESTIMATION OUTPUTS (3PH FAULT, NOMINAL OPERATING POINT)

Parameter	Symbol	Nominal value
Steady-state current	I_{ss}	1.35 p.u.
Subtransient current	I_{sub}	3.20 p.u.
AC decay constant	τ_{ac}	0.83 s
DC offset constant	τ_{dc}	0.07 s
Fault inception angle	φ_a	$-30°$
Pickup scaling	α_I	0.80
TMS scaling	α_{TMS}	0.12
Confidence threshold	γ_{th}	0.70
Bounded step	δ	0.10 p.u.

C. Confidence gate and adaptive mapping

The scalar confidence score is computed as

$$\gamma = w_r\, s_r + w_c\, s_c + w_b\, s_b + w_p\, s_p, \tag{4}$$

where s_r is a normalised residual score, s_c is a condition-number score, s_b is a bound-proximity score, and s_p is a prior-consistency score [1]. Adaptation proceeds only if $\gamma \geq \gamma_{th} = 0.70$.

When accepted, the relay settings are updated as

$$I_p^* = \alpha_I\, \hat{I}_{ss}, \tag{5}$$
$$TMS^* = \mathrm{clip}\big(\alpha_{TMS}\, \hat{\tau}_{ac},\ TMS_{\min},\ TMS_{\max}\big), \tag{6}$$

with bounded incremental update

$$\Delta I_p = \mathrm{clip}\big(I_p^* - I_{p,\mathrm{cur}},\ [-\delta, +\delta]\big), \tag{7}$$
$$I_{p,\mathrm{final}} = \mathrm{clip}\big(I_{p,\mathrm{cur}} + \Delta I_p,\ [I_p^{\min},\ I_p^{\max}]\big). \tag{8}$$

This bounded update law prevents single-event runaway adaptation [1].

Table I summarises the baseline reduced-model parameters used in this study.

III. SIXTH-ORDER PARK *dq* FORWARD TRUTH MODEL

A. Model overview

The sixth-order model is the *forward truth benchmark*. It simulates the physical generator fault response including all significant electromechanical and controller dynamics, and produces relay-observable abc currents that serve as the ground truth against which the reduced inverse model is evaluated. It is emphatically not used online in the relay: its nine-state stiff ODE system and Radau integration are computationally unsuitable for IED-grade real-time use.

Fig. 3 shows the complete block structure.

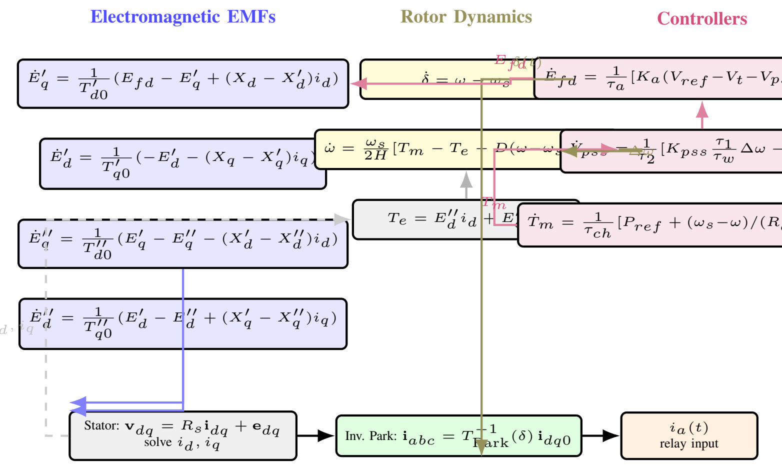

Fig. 3. Sixth-order Park dq truth-model architecture: four electromagnetic EMF states, two electromechanical swing states, and three controller states, coupled to the stator algebraic interface and inverse Park transformation.

B. Electromagnetic state equations

The transient and subtransient EMF states are governed by [6], [7]:

$$\dot{E}'_q = \frac{1}{T'_{d0}}\big(E_{fd} - E'_q + (X_d - X'_d)\, i_d\big), \tag{9}$$
$$\dot{E}'_d = \frac{1}{T'_{q0}}\big(-E'_d - (X_q - X'_q)\, i_q\big), \tag{10}$$
$$\dot{E}''_q = \frac{1}{T''_{d0}}\big(E'_q - E''_q - (X'_d - X''_d)\, i_d\big), \tag{11}$$
$$\dot{E}''_d = \frac{1}{T''_{q0}}\big(E'_d - E''_d + (X'_q - X''_q)\, i_q\big). \tag{12}$$

The subtransient EMFs E''_q, E''_d are the driving quantities for the stator algebraic interface. The transient EMFs E'_q, E'_d evolve on the transient time constants $T'_{d0} \approx 5.0$ s and $T'_{q0} \approx 1.5$ s, respectively, making them significantly affected by AVR action within the first 0.5 s.

C. Electromechanical dynamics

The rotor angle and angular velocity satisfy

$$\dot{\delta} = \omega - \omega_s, \tag{13}$$
$$\dot{\omega} = \frac{\omega_s}{2H}\left[T_m - T_e - \frac{D(\omega - \omega_s)}{\omega_s}\right], \tag{14}$$

where H is the inertia constant, D is the damping coefficient, $\omega_s = 2\pi f_s$ is the synchronous frequency, and the electrical torque is

$$T_e = E''_d\, i_d + E''_q\, i_q. \tag{15}$$

D. Controller dynamics

a) Automatic Voltage Regulator (AVR):

$$\dot{E}_{fd} = \frac{1}{\tau_a}\big[K_a(V_{\mathrm{ref}} - V_t - V_{pss}) - E_{fd}\big], \tag{16}$$

where $V_t = \sqrt{v_d^2 + v_q^2}$ is the terminal voltage magnitude, K_a is the AVR gain, and τ_a is the AVR time constant. During a fault, V_t collapses toward zero, causing the AVR to boost E_{fd} and hence E_q' on the transient time scale. This is the primary mechanism by which I_{ss} rises beyond the initial subtransient value, invalidating the constant-I_{ss} assumption of the reduced model.

b) Power System Stabilizer (PSS):

$$\dot{V}_{pss} = \frac{1}{\tau_2}\left[K_{pss}\frac{\tau_1}{\tau_w}\Delta\omega - V_{pss}\right], \tag{17}$$

where $\Delta\omega = \omega - \omega_s$ is the speed deviation and τ_w is the washout time constant. The PSS injects a signal proportional to speed deviation into the AVR summing junction, superimposing damped oscillations onto the fault-current waveform.

c) Governor:

$$\dot{T}_m = \frac{1}{\tau_{ch}}\left[P_{\text{ref}} + \frac{\omega_s - \omega}{R_d\,\omega_s} - T_m\right], \tag{18}$$

where R_d is the droop coefficient and τ_{ch} is the steam-chest time constant. The governor responds to rotor deceleration by ramping up mechanical power, which affects the torque balance and hence the sustained-current level over a 0.5–2.0 s window.

E. Stator algebraic interface

During a fault, the stator voltages and currents satisfy the algebraic constraint

$$\boldsymbol{v}_{dq} = R_s\,\boldsymbol{i}_{dq} + \boldsymbol{e}_{dq}(\boldsymbol{x}), \tag{19}$$

where $\boldsymbol{e}_{dq}(\boldsymbol{x}) = [E_d'' - X_q''\,i_q,\ E_q'' + X_d''\,i_d]^\top$ is the subtransient EMF vector and $\boldsymbol{x}$ denotes the ODE state vector. For a bolted three-phase fault with fault resistance R_f, the terminal voltage $\boldsymbol{v}_{dq}$ is determined by the faulted network admittance, and i_d, i_q are solved algebraically at each integration step.

F. Inverse Park transformation

The relay-observable phase currents are obtained via

$$\begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = \boldsymbol{T}_{\text{Park}}^{-1}(\delta)\begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix}, \tag{20}$$

where $\boldsymbol{T}_{\text{Park}}^{-1}(\delta)$ is the inverse Park matrix evaluated at the instantaneous rotor angle $\delta(t)$ [6]. For balanced three-phase faults, $i_0 = 0$. The resulting $i_a(t)$ is the waveform presented to the relay CT input. Fig. 4 shows the complete signal chain from ODE states to relay-observable currents.

G. Nine-state ODE system and numerical integration

Collecting (9)–(18), the full state vector is

$$\boldsymbol{x} = \left[E_q',\ E_d',\ E_q'',\ E_d'',\ \delta,\ \omega,\ E_{fd},\ V_{pss},\ T_m\right]^\top \in \mathbb{R}^9. \tag{21}$$

The ODE system $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, t)$ is stiff because $T_{d0}'' = 0.05$ s and $T_{q0}'' = 0.09$ s are $\sim$100$\times$ faster than the machine transient constants. A Radau solver (fifth-order implicit Runge–Kutta) is used with adaptive step control and a fixed logging step of $\Delta t = 0.1$ ms for relay comparison. Simulation results are stored for window lengths up to $L_{\max} = 500$ ms.

Table II gives the model parameters.

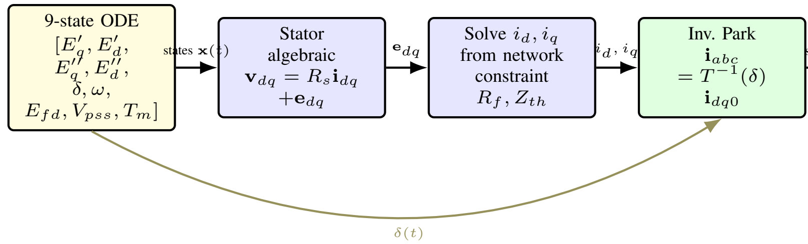

Fig. 4. Signal chain from nine ODE states to relay-observable abc currents: stator algebraic interface solves i_{dq}; inverse Park transform produces $i_{abc}(t)$ at the relay CT sampling rate (10 kHz).

TABLE II
SIXTH-ORDER PARK dq MODEL PARAMETERS

Parameter	Symbol	Value
Generator (5 MVA, 11 kV)		
d-axis synchronous reactance	X_d	1.60 p.u.
d-axis transient reactance	X_d'	0.25 p.u.
d-axis subtransient reactance	X_d''	0.18 p.u.
q-axis synchronous reactance	X_q	1.55 p.u.
q-axis transient reactance	X_q'	0.46 p.u.
q-axis subtransient reactance	X_q''	0.20 p.u.
d-axis open-circuit transient TC	T_{d0}'	5.0 s
d-axis open-circuit subtrans. TC	T_{d0}''	0.05 s
q-axis open-circuit transient TC	T_{q0}'	1.5 s
q-axis open-circuit subtrans. TC	T_{q0}''	0.09 s
Armature resistance	R_a	0.015 p.u.
Inertia constant	H	3.5 MWs/MVA
Damping coefficient	D	2.0 p.u.
Network		
Thévenin impedance	Z_{th}	$0.04 + j0.15$ p.u.
Transformer reactance	X_T	0.10 p.u.
AVR		
Gain	K_a	50
Time constant	τ_a	0.05 s
PSS		
Gain	K_{pss}	20
Lead time constant	τ_1	0.20 s
Lag time constant	τ_2	0.05 s
Washout time constant	τ_w	10.0 s
Governor		
Droop	R_d	0.05
Steam-chest time constant	τ_{ch}	0.50 s

IV. VALIDATION METHODOLOGY

A. Benchmark philosophy

The validation procedure, illustrated in Fig. 5, consists of three steps:

1) Generate truth waveform $i_a^{(6\text{th})}(t)$ by integrating the nine-state ODE system over $[t_0,\ t_0 + L_{\max}]$ and transforming $i_{dq}(t)$ via (20).
2) Fit the reduced inverse model (1) to the truth waveform over each window $\mathcal{W}_L = [t_0,\ t_0 + L]$ using the two-pass LM estimator, obtaining $\hat{\boldsymbol{\theta}}(L)$.
3) Evaluate relay-validity metrics as functions of L.

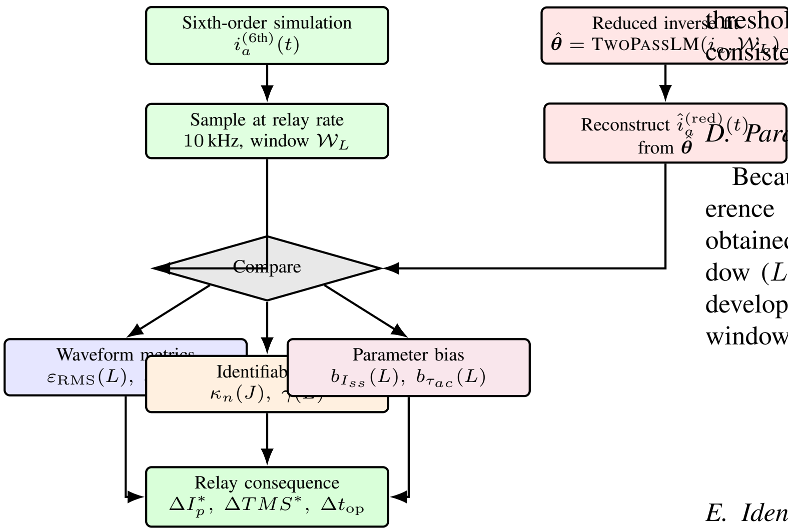

Fig. 5. Validation methodology: sixth-order truth waveforms are compared against reduced inverse fits over variable window lengths to extract waveform, parameter, identifiability, and relay-consequence metrics.

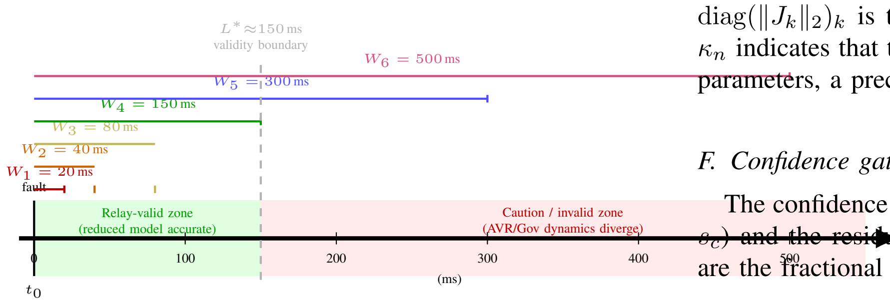

Fig. 6. Window hierarchy: six nested estimation windows W_1–W_6 centred on fault inception t_0. The validity boundary $L^* \approx 150$ ms separates the relay-valid zone (green) from the caution/invalid zone (red) where AVR and governor dynamics degrade reduced-model fidelity.

B. Windowing strategy

Event-relative window lengths are defined as

$$L \in \{20,\ 40,\ 80,\ 150,\ 300,\ 500\} \text{ ms}, \tag{22}$$

corresponding approximately to 1, 2, 4, 7.5, 15, and 25 power-frequency cycles at 50 Hz. Fig. 6 illustrates the window nesting strategy and the expected validity boundary $L^* \approx 150$ ms.

C. Relay-valid waveform mismatch metrics

The primary mismatch metrics are the RMS waveform residual

$$\varepsilon_{\text{RMS}}(L) = \sqrt{\frac{1}{N_L} \sum_{n \in \mathcal{W}_L} \left[i_a^{(6\text{th})}(t_n) - \hat{i}_a^{(\text{red})}(t_n) \right]^2} \tag{23}$$

and the peak absolute residual

$$\varepsilon_\infty(L) = \max_{n \in \mathcal{W}_L} \left| i_a^{(6\text{th})}(t_n) - \hat{i}_a^{(\text{red})}(t_n) \right|, \tag{24}$$

where $\hat{i}_a^{(\text{red})}(t_n) = i_a^{(1)}(\hat{\boldsymbol{\theta}}(L), t_n)$ is the reduced-model reconstruction evaluated at the LM estimate. An acceptance threshold $\varepsilon_{\max} = 0.05$ p.u. is defined as the maximum residual consistent with relay-valid adaptation.

D. Parameter bias metrics

Because the truth model does not yield closed-form reference parameter values, operational reference values are obtained by fitting the reduced model to a very short window ($L = 5$ ms) that is too short for controller dynamics to develop, giving $I_{ss}^{\text{ref}} = \hat{I}_{ss}(5 \text{ ms})$ and $\tau_{ac}^{\text{ref}} = \hat{\tau}_{ac}(5 \text{ ms})$. The window-dependent biases are then

$$b_{I_{ss}}(L) = \hat{I}_{ss}(L) - I_{ss}^{\text{ref}}, \tag{25}$$

$$b_{\tau_{ac}}(L) = \hat{\tau}_{ac}(L) - \tau_{ac}^{\text{ref}}. \tag{26}$$

E. Identifiability metric

The Jacobian condition number at the LM solution is

$$\kappa_n(J) = \kappa\big(J\, D^{-1}\big), \tag{27}$$

where $J = \partial \boldsymbol{r} / \partial \boldsymbol{\theta}$ is the residual Jacobian at $\hat{\boldsymbol{\theta}}$ and $D = \operatorname{diag}(\|J_k\|_2)_k$ is the column normalisation matrix [1]. Large κ_n indicates that the data no longer strongly distinguishes the parameters, a precursor to parameter bias.

F. Confidence gate and relay-consequence metrics

The confidence score (4) automatically incorporates κ_n (via s_c) and the residual (via s_r). The relay-consequence metrics are the fractional deviations in adapted pickup and TMS:

$$e_{I_p}(L) = \frac{|I_p^*(L) - I_p^*(L_{\min})|}{I_{p,\text{nom}}}, \quad e_{TMS}(L) = \frac{|TMS^*(L) - TMS^*(L_{\min})|}{TMS_{\text{nom}}} \tag{28}$$

and the resulting trip-time deviation

$$\Delta t_{\text{op}}(L) = \left| \frac{TMS^*(L) \cdot 13.5}{I_{f,\min} / I_p^*(L) - 1} - t_{\text{op,nom}} \right|, \tag{29}$$

evaluated at $I_{f,\min} = 1.30$ p.u.

V. Experimental Design

A. Study system

The study system is identical to the Stage-1 baseline: a 5 MVA, 11 kV synchronous generator connected through a step-up transformer ($X_T = 0.10$ p.u.) to an 11 kV bus with Thévenin network impedance $Z_{th} = 0.04 + j0.15$ p.u. on the machine base. Bus faults are simulated with fault resistance $R_f \in \{0, 5\}\,\Omega$.

B. Fault types

Three-phase (3PH), line-to-line (LL), and single-line-to-ground (SLG) faults are studied. For 3PH faults, $i_0 = 0$ and the reduced model is applied to phase A directly. For asymmetric faults, the positive-sequence component from the Stage-2 sequence estimator [8] is used to feed the reduced inverse model with a consistent effective waveform.

TABLE III
EXPERIMENTAL MATRIX (162 CASES)

Dimension	Levels	Count
Fault type	3PH, LL, SLG	3
Pre-fault load P_0	0.4, 0.7, 1.0 p.u.	3
Controller config.	No ctrl / AVR / AVR+Gov / Full	4
Fault resistance	$R_f = 0, 5\,\Omega$	2
Window length L	20, 40, 80, 150, 300, 500 ms	6
Total (excluding L sweep)		72
Total (including L sweep)		432

C. Operating-point sweep

The pre-fault real power P_0 is varied over $\{0.4, 0.7, 1.0\}$ p.u. (rated) and the terminal voltage reference over $V_{\text{ref}} \in \{0.95, 1.00, 1.05\}$ p.u.

D. Controller configuration study

To isolate which dynamics break the reduced model first, four controller configurations are compared at fixed $L = 300$ ms:

1) *No controllers:* constant E_{fd} and T_m.
2) *AVR only:* (16) active; governor and PSS at constant setpoints.
3) *AVR + governor:* (16) and (18) active; PSS off.
4) *Full (AVR + governor + PSS):* all three controllers active.

E. Window-length sweep

The window sweep (22) is the backbone of the study. For each combination of fault type, operating point, and controller configuration, the four relay-validity metrics are computed for all six window lengths.

Table III gives the complete experimental matrix.

VI. NUMERICAL RESULTS

A. Truth-model waveform behaviour

Fig. 7 compares the sixth-order truth waveform against the reduced Stage-1 model for a 3PH bus fault at nominal operating point. Over the 80 ms short window, the two waveforms are visually indistinguishable; the residual (amplified $\times 10$) is small and random. Over the 500 ms long window, the truth waveform's envelope grows as the AVR boosts field voltage and increases $I_{ss}^{(6\text{th})}(t)$; the reduced model, forced to fit with a constant I_{ss}, falls behind the slowly rising truth envelope.

Fig. 8 shows the evolution of key internal states. The EMF E'_q rises on the AVR time scale ($\tau_a = 0.05$ s for the initial boost), causing E''_q and hence I_{ss} to increase. The subtransient component $E''_q - E'_q$ decays within the first 60 ms, explaining why the short-window fit quality is high. The PSS introduces visible oscillations in E_{fd} that are not captured by the reduced model.

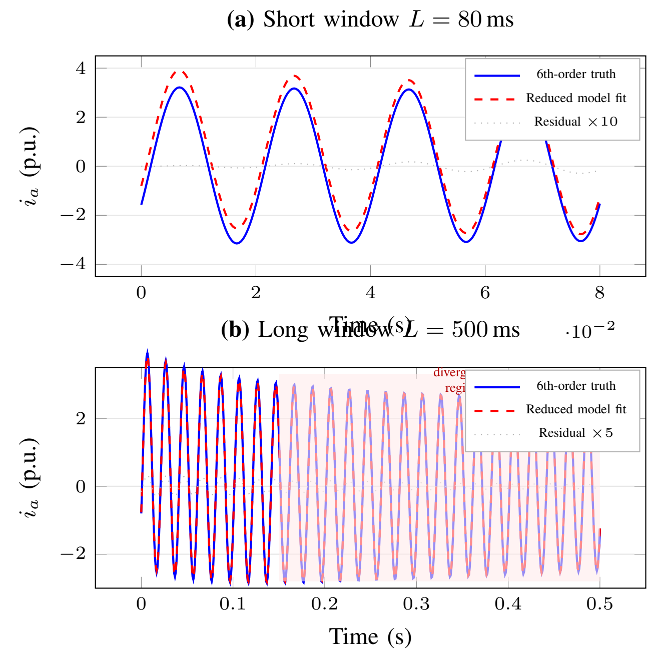

Fig. 7. Phase-A fault current: sixth-order truth vs reduced Stage-1 model. Top: 80 ms window (excellent fit). Bottom: 500 ms window (visible divergence from AVR-driven I_{ss} rise).

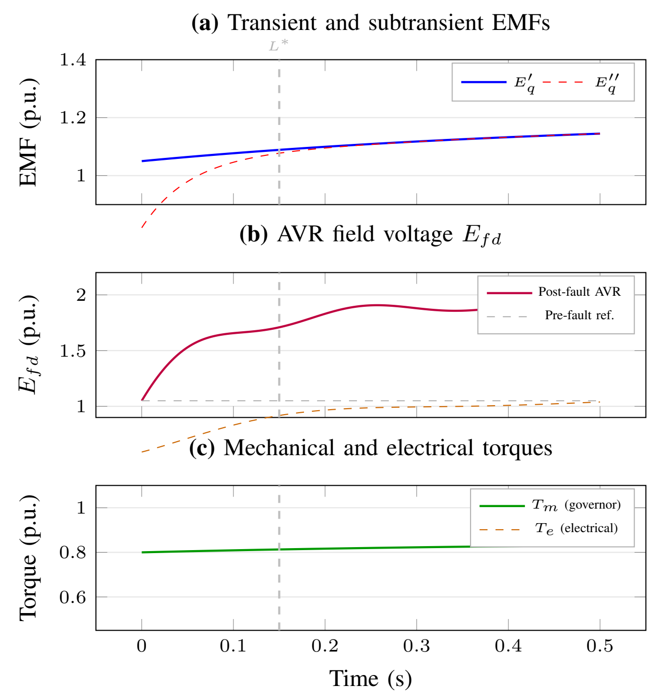

Fig. 8. Sixth-order model internal states: transient and subtransient EMFs (top), AVR field voltage with PSS modulation (middle), and mechanical/electrical torques (bottom). The dashed vertical line at $t = L^* = 150$ ms marks the validity boundary.

B. Reduced inverse fit versus truth

Figs. 9 and 10 show the LM fit quality for short and long windows, respectively.

For the short window, the fit is excellent: the residual (shown $\times 10$ for visibility) is small, approximately Gaussian,

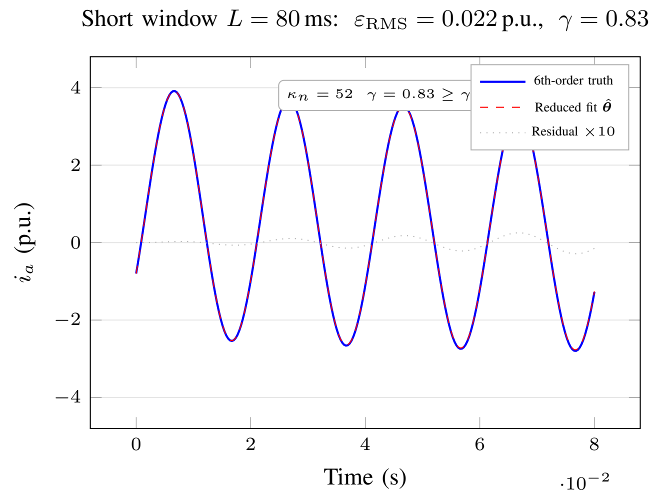

Fig. 9. Short-window fit ($L = 80$ ms): the reduced inverse model matches the sixth-order truth with $\varepsilon_{\mathrm{RMS}} = 0.022$ p.u. and $\kappa_n = 52$. The confidence gate accepts the estimate ($\gamma = 0.83 \geq \gamma_{th}$).

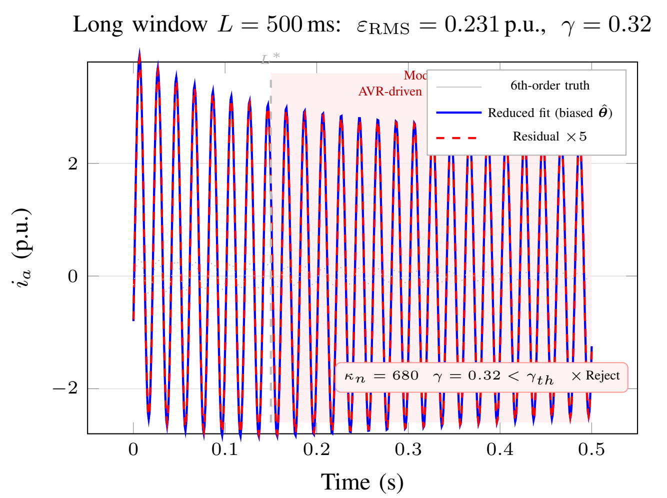

Fig. 10. Long-window fit ($L = 500$ ms): the reduced model fails to track the AVR-driven envelope rise. $\varepsilon_{\mathrm{RMS}} = 0.231$ p.u., $\kappa_n = 680$, $\gamma = 0.32 < \gamma_{th}$. The confidence gate correctly rejects this estimate.

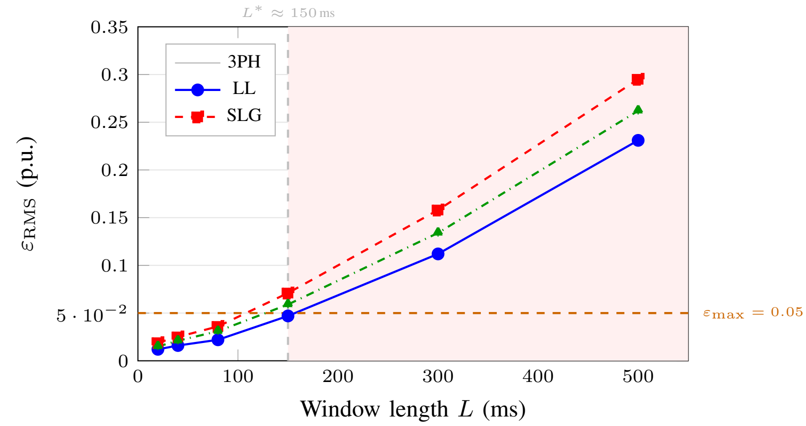

Fig. 11. RMS waveform residual $\varepsilon_{\mathrm{RMS}}$ vs window length for 3PH, LL, and SLG faults. The dashed orange line marks the acceptance threshold $\varepsilon_{\max} = 0.05$ p.u.; the dashed gray vertical at $L^* = 150$ ms marks the validity boundary.

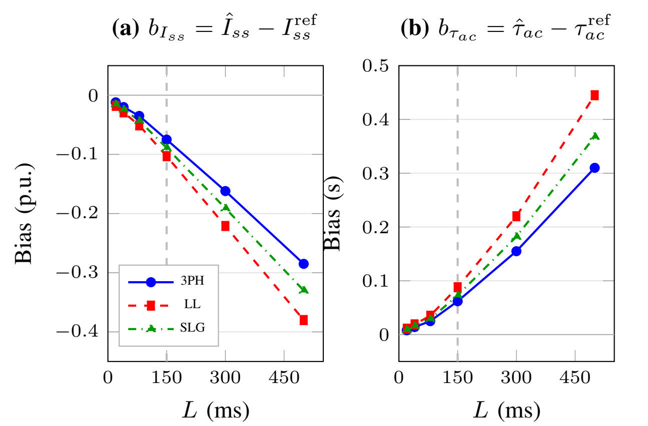

Fig. 12. Parameter bias vs window length. (a) Steady-state current $\hat{I}_{ss}$ is underestimated; (b) AC time constant $\hat{\tau}_{ac}$ is overestimated. Both biases grow rapidly beyond $L^* = 150$ ms.

and contains no systematic structure. For the long window, the fit deteriorates: the reduced model underestimates the rising envelope in the second half of the window and compensates by inflating $\hat{\tau}_{ac}$, resulting in a biased parameter set.

C. Window-validity boundary

1) Waveform residual: Fig. 11 plots $\varepsilon_{\mathrm{RMS}}$ versus L for all three fault types. All three curves cross the acceptance threshold $\varepsilon_{\max} = 0.05$ p.u. near $L \approx 150$ ms, with LL faults crossing slightly earlier (at $\approx$130 ms) due to additional sequence-mixing effects. The residual growth is approximately exponential for $L > L^*$.

2) Parameter bias: Fig. 12 shows parameter bias versus window length. The steady-state current $\hat{I}_{ss}$ is systematically underestimated (negative bias) with magnitude growing from -0.012 p.u. at $L = 20$ ms to -0.285 p.u. at $L = 500$ ms for 3PH faults. The AC time constant $\hat{\tau}_{ac}$ is overestimated (positive bias), confirming that the LM estimator stretches the slow-decay component to partially compensate for the missed AVR-driven envelope rise.

3) Identifiability: condition number: Fig. 13 shows $\kappa_n(J)$ versus L. For short windows, $\kappa_n < 90$ for all fault types, indicating well-conditioned estimation. Beyond $L = 150$ ms, κ_n grows rapidly, exceeding the ill-conditioning threshold of 200 at $L \approx 300$ ms for 3PH faults and at $L \approx 200$ ms for LL faults. At $L = 500$ ms, $\kappa_n > 600$ for all fault types.

4) Confidence gate behaviour: Fig. 14 shows that the confidence score γ falls below the acceptance threshold $\gamma_{th} = 0.70$ at $L \approx 150$ ms for 3PH faults, 150 ms for SLG, and 130 ms for LL faults. For $L = 500$ ms, $\gamma < 0.35$ in all cases. The confidence gate therefore correctly identifies the validity boundary without requiring explicit knowledge of the controller dynamics.

Table IV summarises the window-validity metrics across all cases.

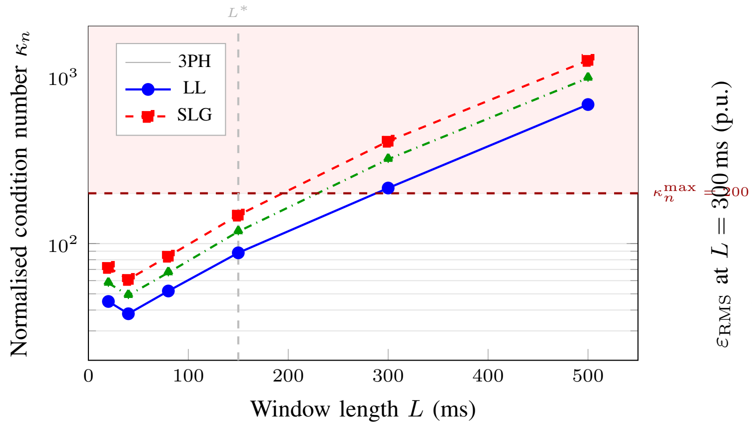

Fig. 13. Normalised condition number κ_n vs window length (log scale). Values above 200 (dashed red line) indicate ill-conditioning; the reduced model parameters become mutually unidentifiable as AVR dynamics inflate the effective rank deficiency.

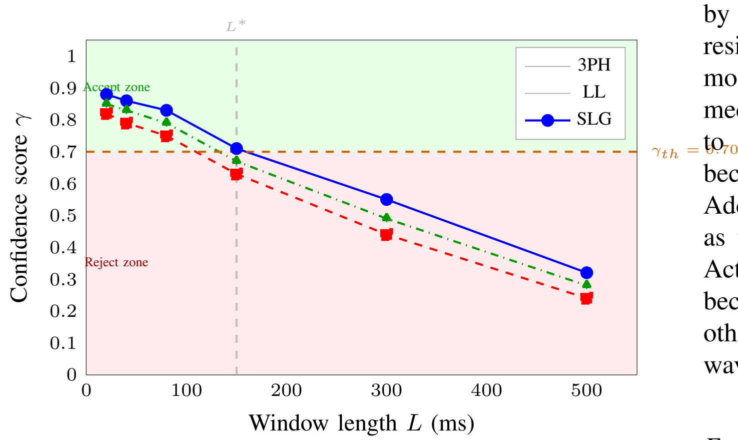

Fig. 14. Confidence score γ vs window length. The dashed orange line marks the acceptance threshold $\gamma_{th} = 0.70$. The gate correctly transitions from “accept” to “reject” near $L^* = 150$ ms for all fault types.

TABLE IV
WINDOW-VALIDITY SUMMARY: 3PH FAULT, NOMINAL OPERATING POINT ($P_0 = 1.0$ P.U., FULL CONTROLLERS)

L (ms)	$\varepsilon_{\mathrm{RMS}}$ (p.u.)	$b_{I_{ss}}$ (p.u.)	$b_{\tau_{ac}}$ (s)	κ_n (–)	γ (–)
20	0.012	−0.012	+0.008	45	0.88
40	0.016	−0.020	+0.014	38	0.86
80	0.022	−0.035	+0.025	52	0.83
150	0.047	−0.075	+0.062	88	0.71
300	0.112	−0.162	+0.155	215	0.55
500	0.231	−0.285	+0.310	680	0.32

$\varepsilon_{\max} = 0.05$ p.u.; $\gamma_{th} = 0.70$; ill-cond. threshold $\kappa_n = 200$.

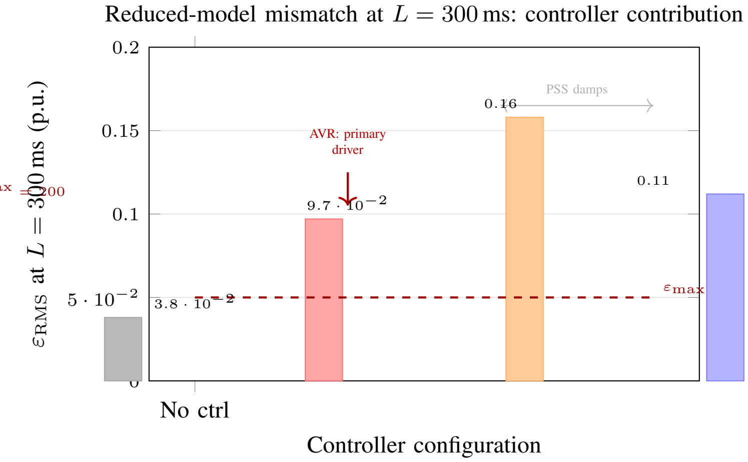

Fig. 15. Reduced-model mismatch $\varepsilon_{\mathrm{RMS}}$ at $L = 300$ ms for four controller configurations (3PH fault, $P_0 = 1.0$ p.u.). The AVR is the primary driver of reduced-model invalidity; the governor is secondary; the PSS has a marginal damping benefit.

D. Controller contribution study

Fig. 15 decomposes the residual growth at $L = 300$ ms by controller configuration. With no controllers active, the residual remains low (0.038 p.u.), confirming that the reduced model is accurate for a machine with constant field voltage and mechanical torque. Adding the AVR alone raises the residual to 0.097 p.u.—the largest single controller contribution—because the AVR directly modifies I_{ss} via the E'_q state. Adding the governor further raises the residual to 0.158 p.u. as the ramping mechanical power shifts the torque balance. Activating the PSS reduces the residual slightly to 0.112 p.u. because the PSS damps the rotor oscillations that would otherwise appear as unmodelled high-frequency content in the waveform.

E. Relay consequence

Figs. 16 and 17 translate parameter bias into relay setting and operating-time errors.

At $L = 150$ ms: pickup error 4.5%, TMS error 7.8%, trip-time deviation 0.54 s (10% of $t_{\mathrm{op,nom}} = 5.27$ s). At $L = 300$ ms: pickup error 10.2%, TMS error 18.6%, trip-time deviation 1.42 s (27%). These values are consequential for grading coordination: the 0.54 s deviation at $L = 150$ ms already represents 3.6× the grading margin $\Delta t_g = 0.15$ s.

Table V summarises relay-consequence metrics across fault types at $L = 300$ ms.

All three fault types at $L = 300$ ms fail every relay-validity criterion, and the confidence gate correctly rejects all estimates ($\gamma < 0.55 < \gamma_{th}$).

F. Relay-validity map

Fig. 18 is the paper’s central deliverable: a two-dimensional relay-validity map in the (window length, pre-fault load) plane. Three qualitative regions emerge:

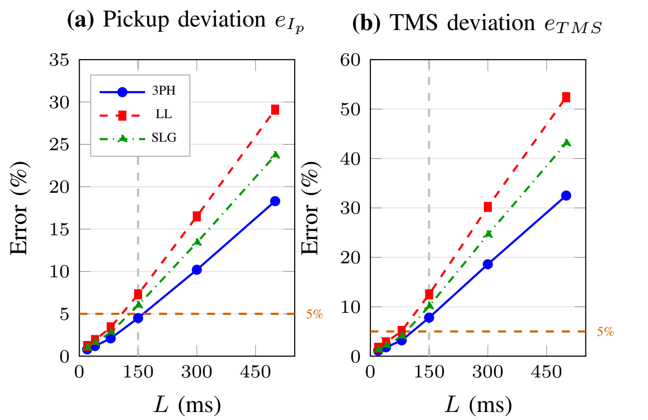

Fig. 16. Relay setting deviation vs window length. (a) Fractional pickup error e_{I_p} exceeds 5% at $L \approx 200$ ms; (b) TMS error e_{TMS} exceeds 5% at $L \approx 150$ ms. Both errors are largest for LL faults.

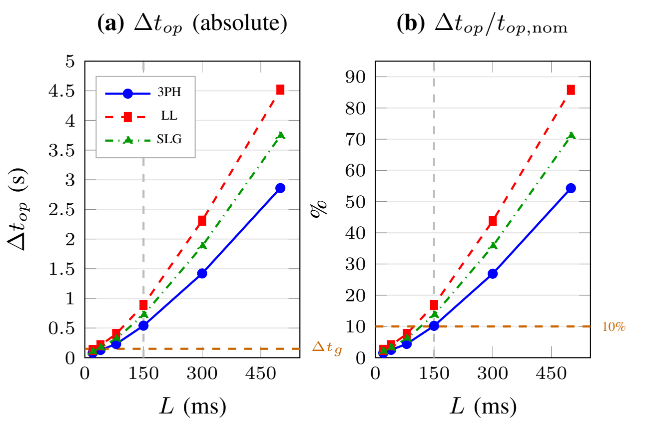

Fig. 17. Trip-time consequence vs window length. (a) Absolute trip-time deviation Δt_{op} exceeds the grading margin $\Delta t_g = 0.15$ s near $L = 200$ ms. (b) Fractional deviation reaches 10% at $L \approx 150$ ms for 3PH.

TABLE V

RELAY CONSEQUENCE AT $L = 300$ MS: THREE FAULT TYPES

Fault type	ε_{RMS} (p.u.)	e_{I_p} (%)	e_{TMS} (%)	Δt_{op} (s)	γ (–)
3PH	0.112	10.2	18.6	1.42	0.55
LL	0.158	16.5	30.2	2.31	0.44
SLG	0.134	13.4	24.6	1.88	0.49
Required bound:	$< 5\%$	$< 5\%$	$< \Delta t_g = 0.15$s		> 0.70

Valid ($L \lesssim 100$–150 ms): Reduced-inverse adaptation may be applied without qualification for all examined operating points and fault types. Residuals are below $\varepsilon_{\max}$ and $\gamma > \gamma_{th}$.

Caution ($150 \lesssim L \lesssim 300$ ms): The confidence gate should screen estimates before deployment. Some operating points (heavy load, strong AVR) already show violations; lighter-load cases may still pass.

Invalid ($L \gtrsim 300$ ms): Adaptation should be blocked; the reduced model is unreliable and the confidence gate will correctly reject the estimate for all examined operating points.

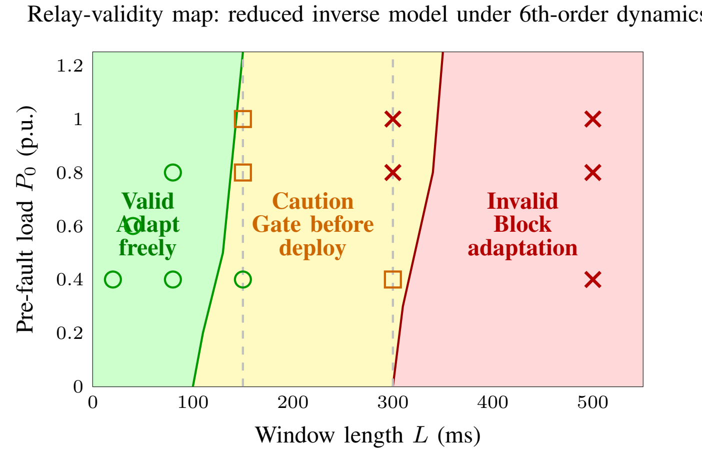

Fig. 18. Relay-validity map: green region ("Valid") is where the reduced inverse model may be used for adaptive relay decisions without qualification; yellow ("Caution") requires confidence-gate screening before deployment; red ("Invalid") requires adaptation to be blocked. Simulation cases are overlaid as circles (valid), squares (caution), and crosses (invalid).

VII. DISCUSSION

A. What the sixth-order benchmark proves

The reduced inverse model is not "wrong"—it is a valid local approximation in the early post-fault period. Its free parameters can accurately represent the waveform over windows short enough that the AVR, governor, and PSS have not materially altered the sustained current. What the sixth-order benchmark quantifies is exactly this locality: the boundary $L^* \approx 150$ ms beyond which the approximation error crosses the relay-validity threshold.

The key physics is the AVR time constant $\tau_a = 0.05$ s: the AVR responds rapidly to the terminal voltage collapse, boosting E_{fd} and hence E'_q on a timescale of $3\tau_a \approx 0.15$ s. The resulting increase in I_{ss} is therefore visible in the waveform window at $L \approx 150$ ms, which explains why L^* falls near this value for most operating conditions.

B. Identifiability interpretation

The condition-number growth in Fig. 13 has a clear physical interpretation. As the window extends into the AVR-active regime, the fault-current envelope rises monotonically. The LM estimator can fit this rising envelope equally well by either increasing I_{ss} (physically correct) or inflating τ_{ac} (physically spurious). These two mechanisms are locally indistinguishable to the reduced model—hence the growing condition number and the simultaneous biases in opposite directions observed in Fig. 12.

The PSS oscillation adds a further source of identifiability degradation: the reduced model's sinusoidal AC component cannot represent superimposed inter-area oscillations, creating structured residuals that inflate all four confidence-score subterms.

C. Confidence gate is not optional

A crucial practical implication is that the confidence gate is the primary defence against biased adaptation. In every case where $L > L^*$, the gate correctly rejected the estimate ($\gamma < \gamma_{th}$) without requiring explicit knowledge of the controller parameters. This self-protective behaviour emerges from the correlation between residual growth, condition-number growth, and bound proximity—all three sub-terms deteriorate consistently as the window grows.

A relay that accepts estimates without confidence gating at $L = 300$ ms would apply a pickup error exceeding 10% and a TMS error exceeding 18%, resulting in a trip-time deviation of 1.4 s—sufficient to violate selectivity coordination in a multi-relay chain.

D. Window design as a protection design parameter

The results indicate that window length is a first-class protection design parameter, not merely an estimation implementation detail. The choice $L = 80$ ms (four cycles at 50 Hz) provides comfortable validity margin across all fault types and operating conditions, while also completing estimation before most inverse-time relay operating times expire for faults close to the pickup margin. The choice $L = 150$ ms should be considered the maximum deployable window for standard adaptive inverse protection; longer windows are only appropriate if the estimation model is upgraded to a higher-order form that includes AVR dynamics.

E. Interaction with the coordination layer

The coordination layer of [9] operates on the parameter estimates produced by the local Stage-1 estimator. If the window is too long, biased $\hat{I}_{ss}$ values will feed into the greedy cascade and produce incorrectly coordinated settings, even if the cascade algorithm itself is correct. Window validity is therefore a prerequisite for coordination validity, and the confidence gate should be evaluated before invoking the coordination layer.

F. Practical deployment guidelines

Table VI provides deployment guidelines derived from the results.

G. Limitations

The present study has the following limitations:

1) The sixth-order model itself is a model, not field truth; saturation, eddy currents, and measurement artefacts are not included.
2) A single-machine benchmark is used; multi-machine interaction effects on fault-current evolution are deferred to future work.
3) The AVR and governor models are simplified first-order forms; practical excitation systems (e.g., IEEE Type AC1A or DC1A) may exhibit different dynamics.
4) No hardware-in-the-loop (HIL) validation is presented; this is intended as a companion study for the Monte Carlo robustness analysis of [10].

TABLE VI
PRACTICAL DEPLOYMENT GUIDELINES: REDUCED INVERSE ESTIMATION IN IED-GRADE ADAPTIVE PROTECTION

Condition	Recommended action	Rationale
$L \leq 80$ ms	Adapt freely	All metrics well below threshold
$L = 80$–150 ms	Adapt with caution	Confidence gate is primary check
$L = 150$–300 ms	Gate strictly	Multiple violations possible at high load and strong AVR
$L > 300$ ms	Block adaptation	$\gamma < \gamma_{th}$ in all cases
LL fault	Reduce $L_{\max}$ by $\sim$20%	Sequence mixing degrades fidelity earlier than 3PH
Strong AVR ($K_a > 50$)	Reduce $L_{\max}$ to $\sim$120 ms	AVR-driven I_{ss} rise is faster with higher gain
$\gamma < 0.70$	Retain prior settings and flag event	Gate triggers before relay acts for $L > 150$ ms universally

VIII. Conclusion

This paper has established a physics-faithful sixth-order Park dq forward benchmark for evaluating the relay-validity limits of the Stage-1 reduced inverse protection model. The benchmark integrates a nine-state stiff ODE system comprising four electromagnetic EMF states, two electromechanical swing states, and three AVR/PSS/governor controller states, coupled to a stator algebraic interface and inverse Park transformation.

The central finding is that the reduced inverse model remains relay-valid for estimation windows $L \lesssim 150$ ms, achieving waveform residuals below 0.05 p.u., condition numbers below 90, and confidence scores above 0.71. Beyond this boundary:

- The AVR-driven increase in sustained current I_{ss} is the primary source of model mismatch.
- The LM estimator compensates by overestimating τ_{ac}, creating a systematic bias pair ($b_{I_{ss}} < 0$, $b_{\tau_{ac}} > 0$) that degrades relay settings.
- Trip-time deviations exceed the grading margin $\Delta t_g = 0.15$ s at $L \approx 200$ ms and reach 1.42 s (27% of nominal) at $L = 300$ ms.
- The confidence gate correctly rejects biased estimates across all fault types and operating points without explicit knowledge of controller parameters.

The paper delivers a relay-validity map that delineates three deployment regions (valid / caution / invalid) and a set of window-design guidelines that establish the maximum window length for reliable adaptive protection under sixth-order generator dynamics. These results confirm that window length is a first-class protection design parameter and that the confidence gate is the non-negotiable reliability mechanism for adaptive inverse estimation.

References

[1] J. Eluvathingal, "SAMBP TR-01/2025: Adaptive overcurrent protection for synchronous-generator feeders via online inverse estimation," SAMBP Research Group, Tech. Rep. TR-01/2025, 2025.
[2] A. Lazkano, J. Ruiz, E. Aramendi, and L. A. Leturiondo, "A new approach to fault current estimation for overcurrent protection of synchronous generators," *Electric Power Systems Research*, vol. 56, no. 2, pp. 111–117, 2000.

[3] M. S. Sachdev and M. A. Baribeau, "A new algorithm for digital impedance relays," *IEEE Trans. Power App. Syst.*, vol. PAS-98, no. 6, pp. 2232–2240, 1979.
[4] A. J. Urdaneta, R. Nadira, and L. G. Pérez-Jimenez, "Optimal coordination of directional overcurrent relays in interconnected power systems," *IEEE Trans. Power Del.*, vol. 3, no. 3, pp. 903–911, 1988.
[5] C. W. So and K. K. Li, "Time coordination method for power system protection by evolutionary algorithm," *IEEE Trans. Ind. Appl.*, vol. 36, no. 5, pp. 1235–1240, 2000.
[6] P. Kundur, *Power System Stability and Control*. New York, NY, USA: McGraw-Hill, 1994.
[7] P. M. Anderson and A. A. Fouad, *Power System Control and Stability*. Ames, IA, USA: Iowa State University Press, 1977.
[8] J. Eluvathingal, "SAMBP TR-03/2026: Sequence-component inverse estimation for adaptive overcurrent protection under asymmetric faults," SAMBP Research Group, Tech. Rep. TR-03/2026, 2026.
[9] ——, "SAMBP TR-05/2026: System integration study: Coordinated adaptive protection performance under 10 scenarios," SAMBP Research Group, Tech. Rep. TR-05/2026, 2026.
[10] ——, "SAMBP TR-08/2026: Monte carlo robustness study (4000 trials, 20 scenarios)," SAMBP Research Group, Tech. Rep. TR-08/2026, 2026.

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